

1	2	3	4	5	6	7	8	
SAM CPU replacement board, draft 0.2, drawn from docs/ARCHITECTURE.md. Licence: CERN–OHL–S v2. A Raspberry Pi 4 on the 40–pin header J21 runs PinMAME + ppuc (HDMI stays on the Pi). The RP2354B drives the original IO power driver board on J9 and scans the switches. Connector names and pinouts follow the original CPU/Sound board 520–5246–00. Each connector sits on the same page as the circuit that drives or reads it. Signals between pages use global labels; the numbers in brackets next to a label list the other pages it goes to.								
A								A
B	Power Power: +5 V mux, external +12 V, 3.3 V, 4.5 V, monitors File: power.kicad_sch		MCU and Pi RP2354B SAM IO controller and Raspberry Pi 4 header File: mcu.kicad_sch		IO bus J9: bus to the IO power driver board (IO board J1) File: io_bus.kicad_sch			B
C	Switch columns Switch matrix columns: 74HC595, strobe drivers, J1 File: sw_columns.kicad_sch		Switch rows 1–8 Switch matrix returns 1–8: J6, LM339 front end, 74HC165 File: sw_rows_1.kicad_sch		Switch rows 9–16 Switch matrix returns 9–16: J12, LM339 front end, 74HC165 File: sw_rows_9.kicad_sch			C
D	Dedicated switches 1–16 Dedicated switches D1–D16 (J2, J3) and coin door memory protect File: sw_dedicated_1.kicad_sch		Dedicated switches 17–24 Dedicated switches D17–D24 (J13) and DIP switches File: sw_dedicated_2.kicad_sch		Display and GI J5 original DMD driver and J18 GI dimmer header File: display_gi.kicad_sch			D
E	Audio Audio: PCM5102A, TDA7297 stereo + subwoofer (+12 V) File: audio.kicad_sch							E
F								F
1	2	3	4	5	6	7	8	

Stern SAM CPU PPUC (CERN–OHL–S v2)

Sheet: /
File: sam_cpu.kicad_sch

Title: SAM CPU replacement board

Size: A3

Date: 2026–10–06

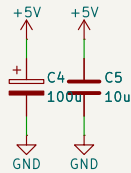
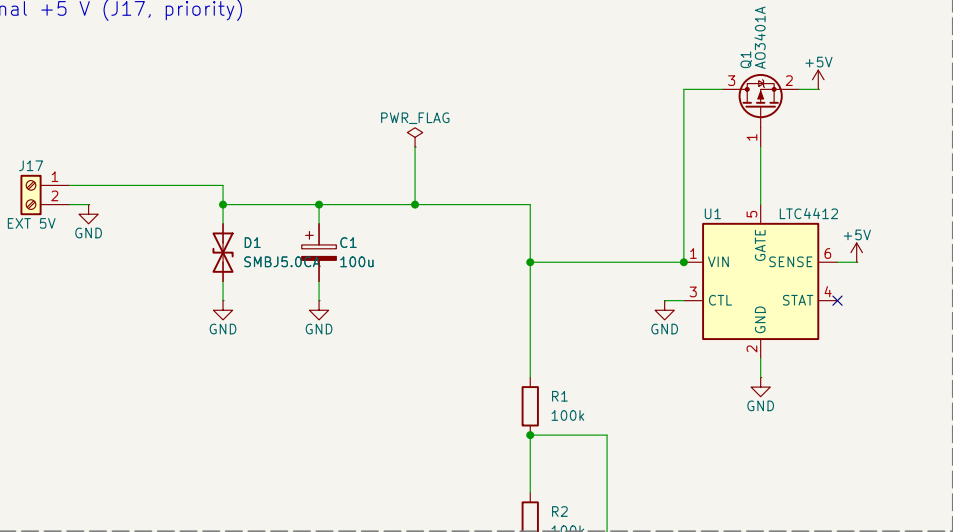
Rev: 0.2

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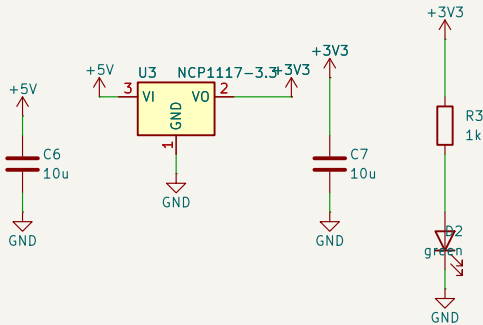
Id: 1/12

J11 keeps the original CPU board connector (IO board J16 harness) for +5 V and ground only: its +−12 V pins are not used.
The audio amplifiers run from an external +12 V supply on J23 (about 3 A at full volume). F1 + D5 blow the fuse on reverse polarity.
U3 NCP1117 (as in the RP2350 hardware guide) is stable with ceramic capacitors. All regulator capacitors are 0805 or larger for low DC−bias derating.
External +5 V on J17 has priority: U2 turns the J11 path off whenever +5V_EXT is present (EXT_PRESENT high).
LTC4412 + AO3401A ideal diodes stand in for the TPS2121 of ARCHITECTURE.md 9.3 (library part, about 4 A per path).
JP1 (bridged by default) disconnects the J11 +5 V completely; leaving J17 unplugged forces J11.

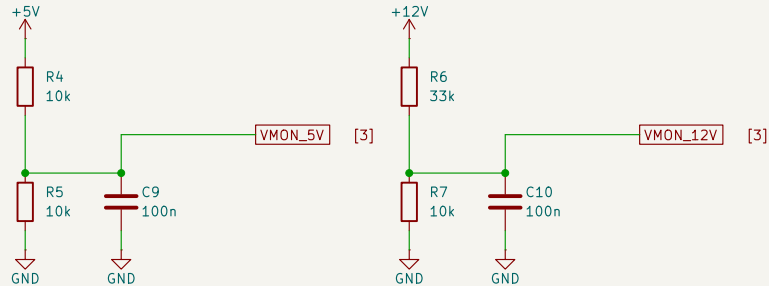
External +5 V (J17, priority)



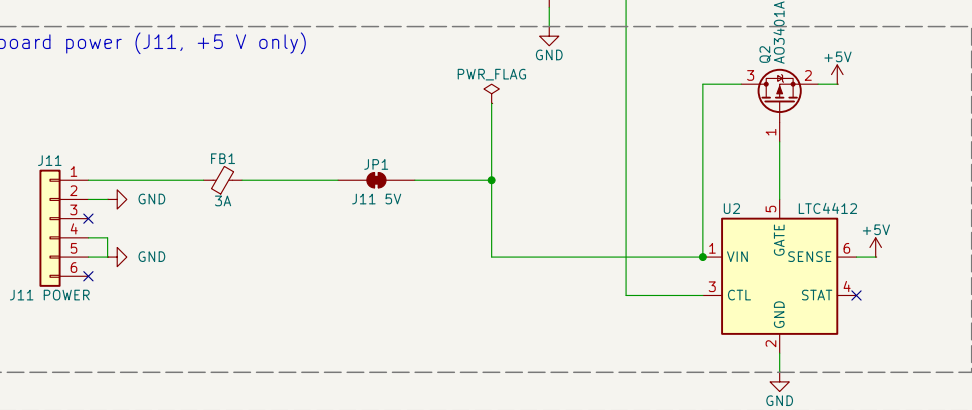
3.3 V



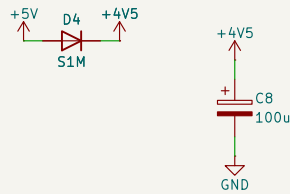
Supply monitors (RP2354B ADC)



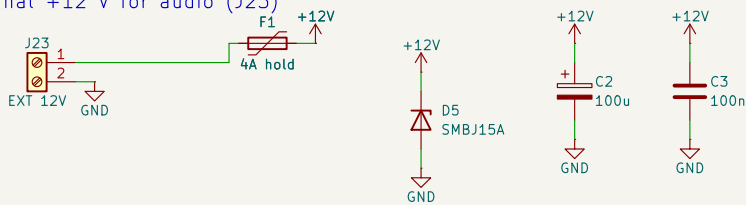
IO board power (J11, +5 V only)



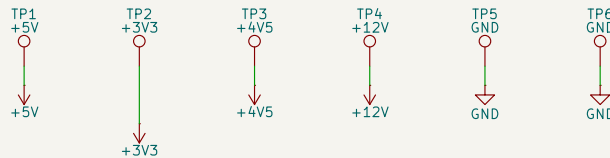
4.5 V for the switch matrix



External +12 V for audio (J23)



Test points



Stern SAM CPU PPUC (CERN−OHL−S v2)

Sheet: /Power/
File: power.kicad_sch

Title: Power: +5 V mux, external +12 V, 3.3 V, 4.5 V, monitors

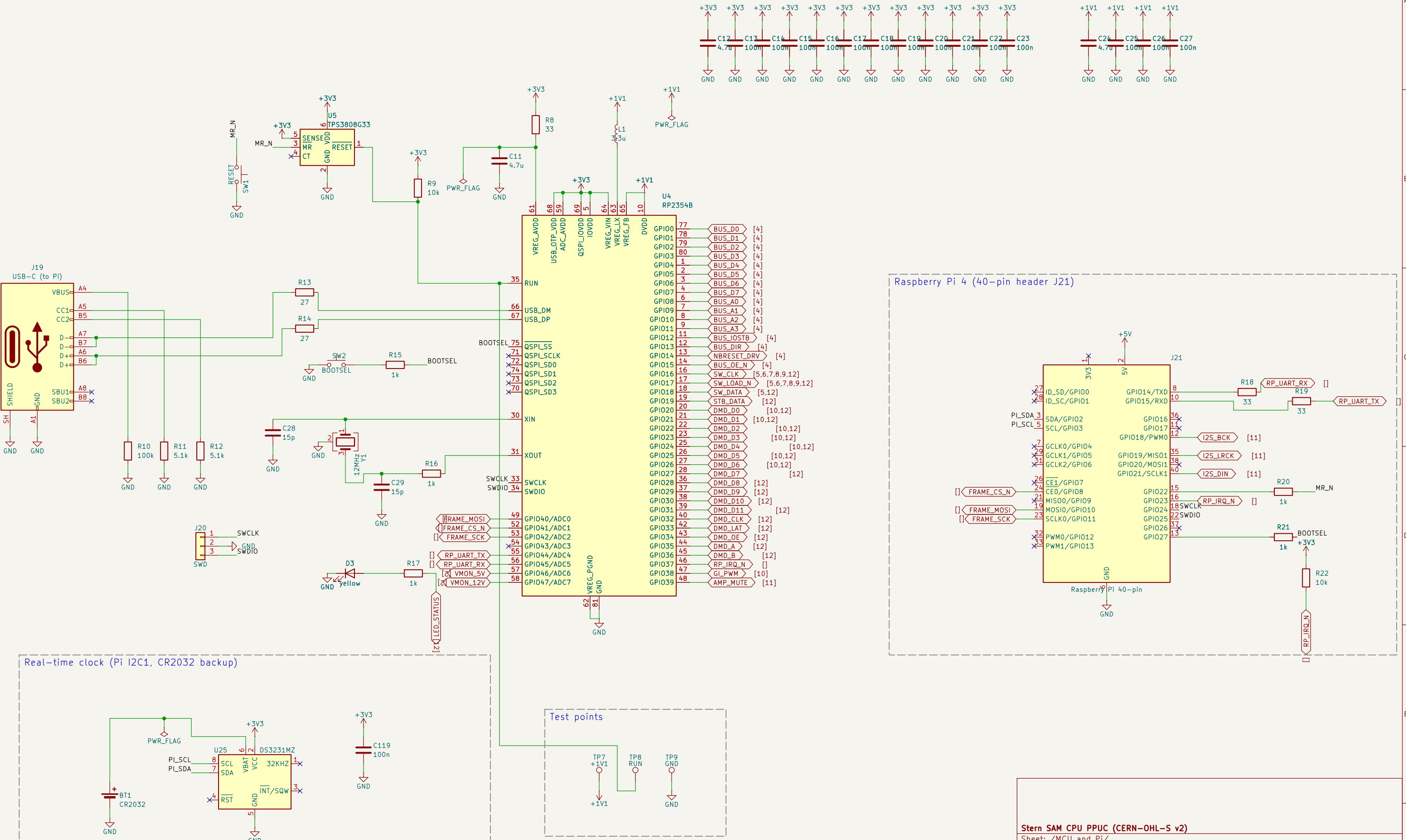
Size: A3 Date: 2026−10−06

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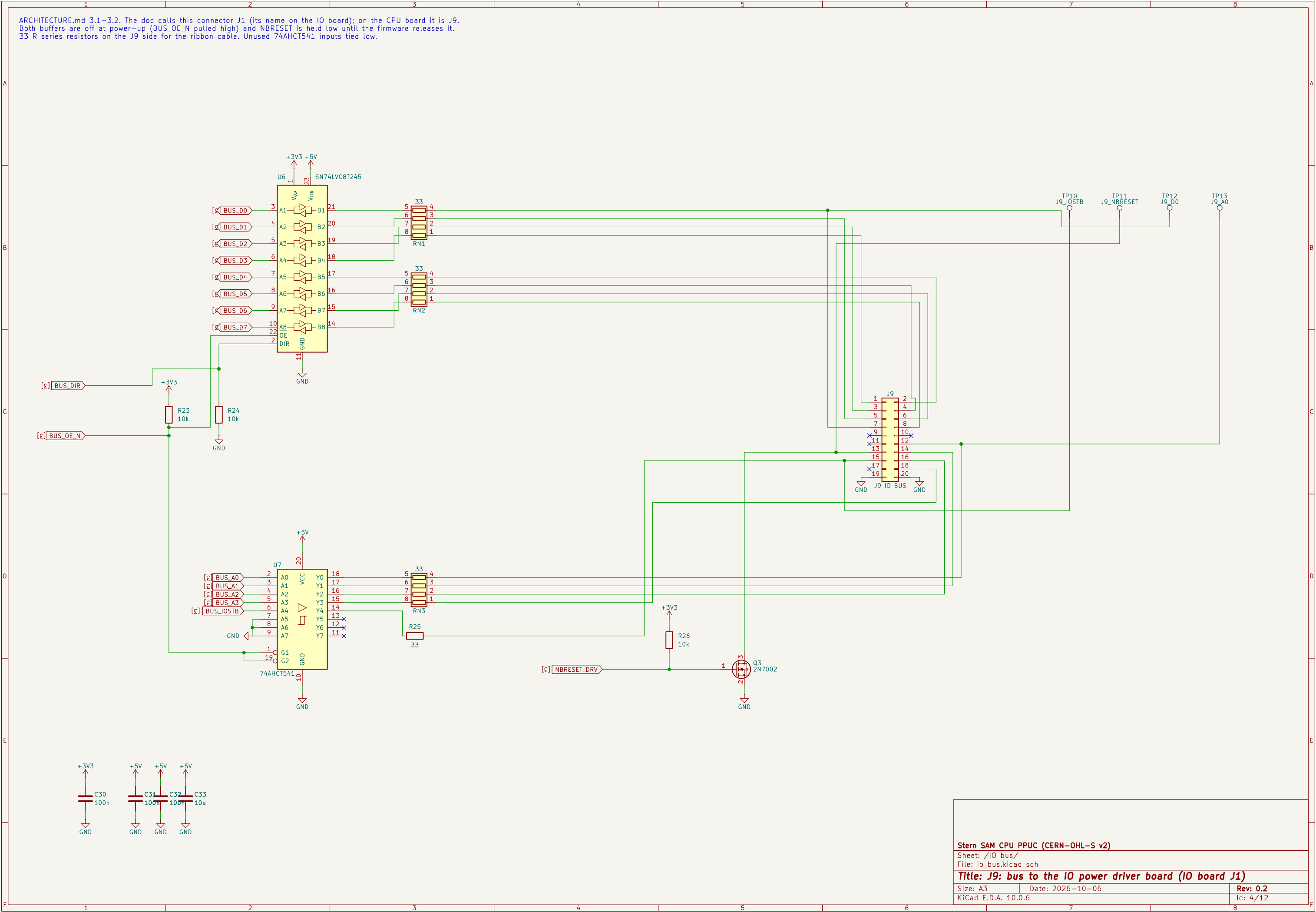
Rev: 0.2

Id: 2/12

GPIO map: ARCHITECTURE.md 9.2. PIO0 bus on GPIO0–15, PIO1 switch chain on GPIO16–20, PIO2 drives two HD DMD panels on GPIO20–36 (data 20–31, CLK, LAT, OE, A, B); the original DMD (J5) shares GPIO20–26. Frames from the Pi on SPI1 (GPIO40–42), UART0 on GPIO44–45, ADC monitors on GPIO46–47, GPIO43 spare. Coin door, status LED and GL_SPARE moved onto the switch chain. Core supply from the internal switching regulator: VREG_LX → L1 → DVDD (1.1 V); VREG_AVDD through 33 R / 4.7 uF. Check against the RP2350 hardware guide. Raspberry Pi 4 on the 40-pin header J21 (HDMI stays on the Pi). The Pi is powered from the board +5 V; never plug the Pi USB–C supply at the same time. U25 DS3231MZ real-time clock on the Pi I2C1 bus (Linux: dtoverlay=i2c-rtc.ds3231), backed by a CR2032 in BT1.



ARCHITECTURE.md 3.1–3.2. The doc calls this connector J1 (its name on the IO board); on the CPU board it is J9. Both buffers are off at power-up (BUS_OE_N pulled high) and NBRESET is held low until the firmware releases it. 33 R series resistors on the J9 side for the ribbon cable. Unused 74AHCT541 inputs tied low.



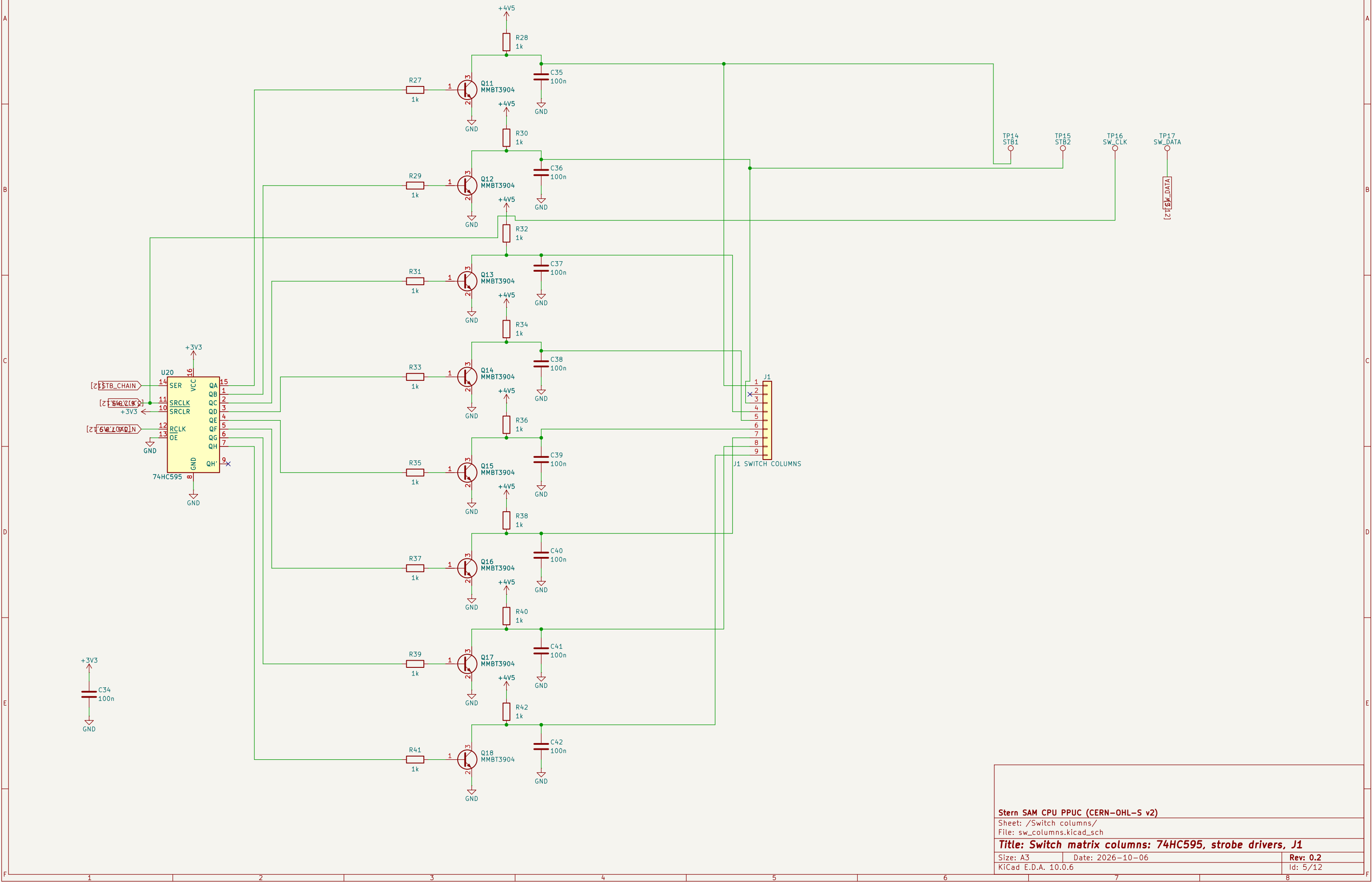
Stern SAM CPU PPUC (CERN-OHL-S v2)

Sheet: /IO bus/
File: io_bus.kicad_sch

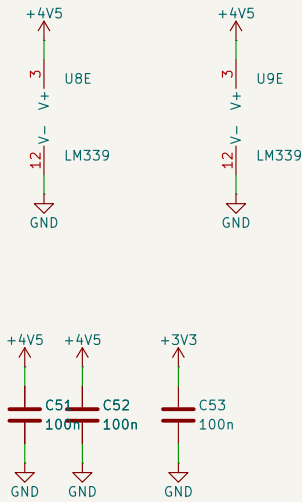
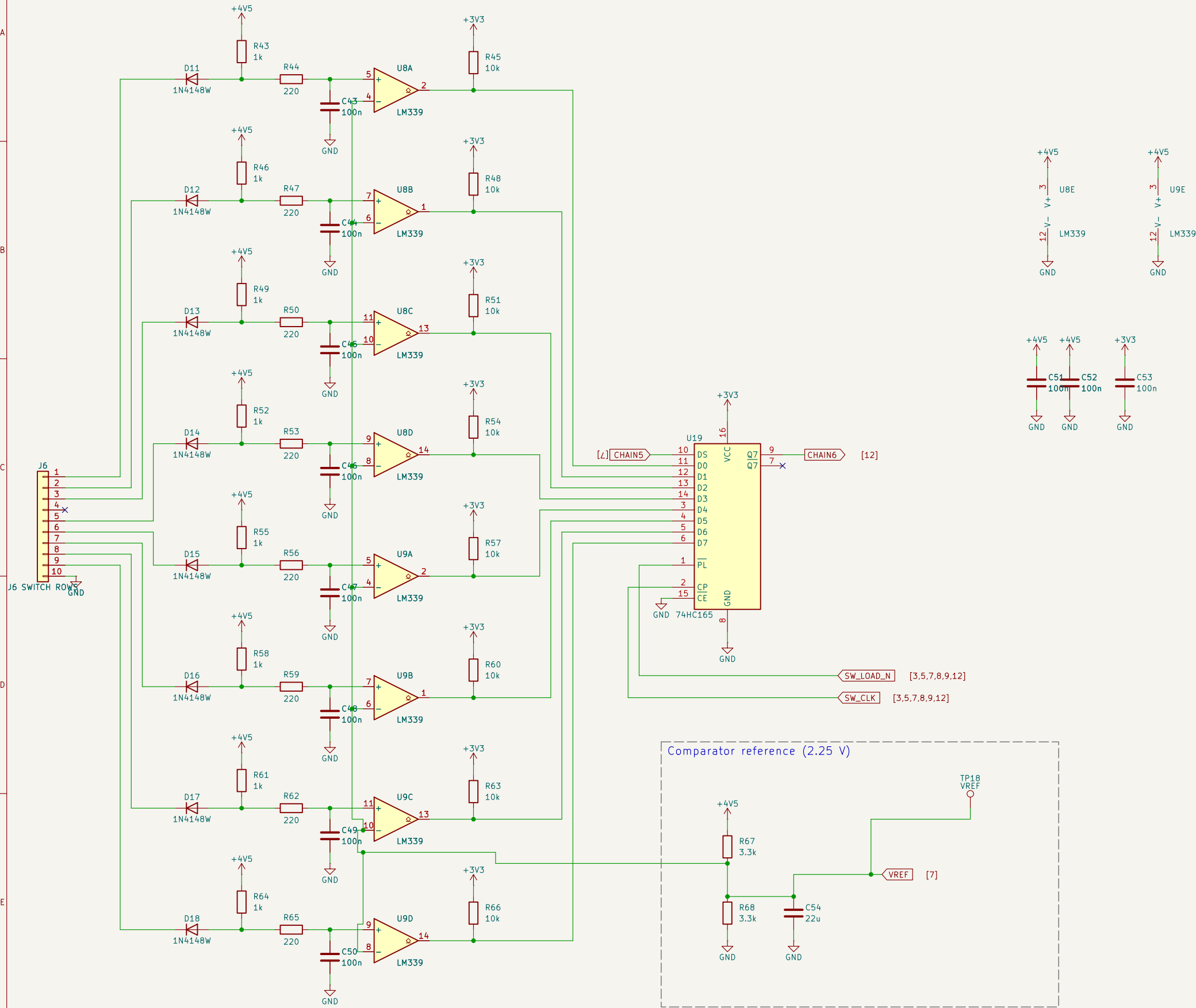
Title: J9: bus to the IO power driver board (IO board J1)

Size: A3	Date: 2026-10-06	Rev: 0.2
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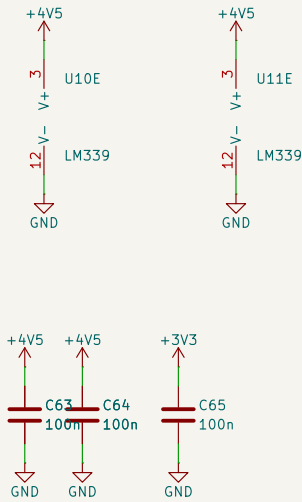
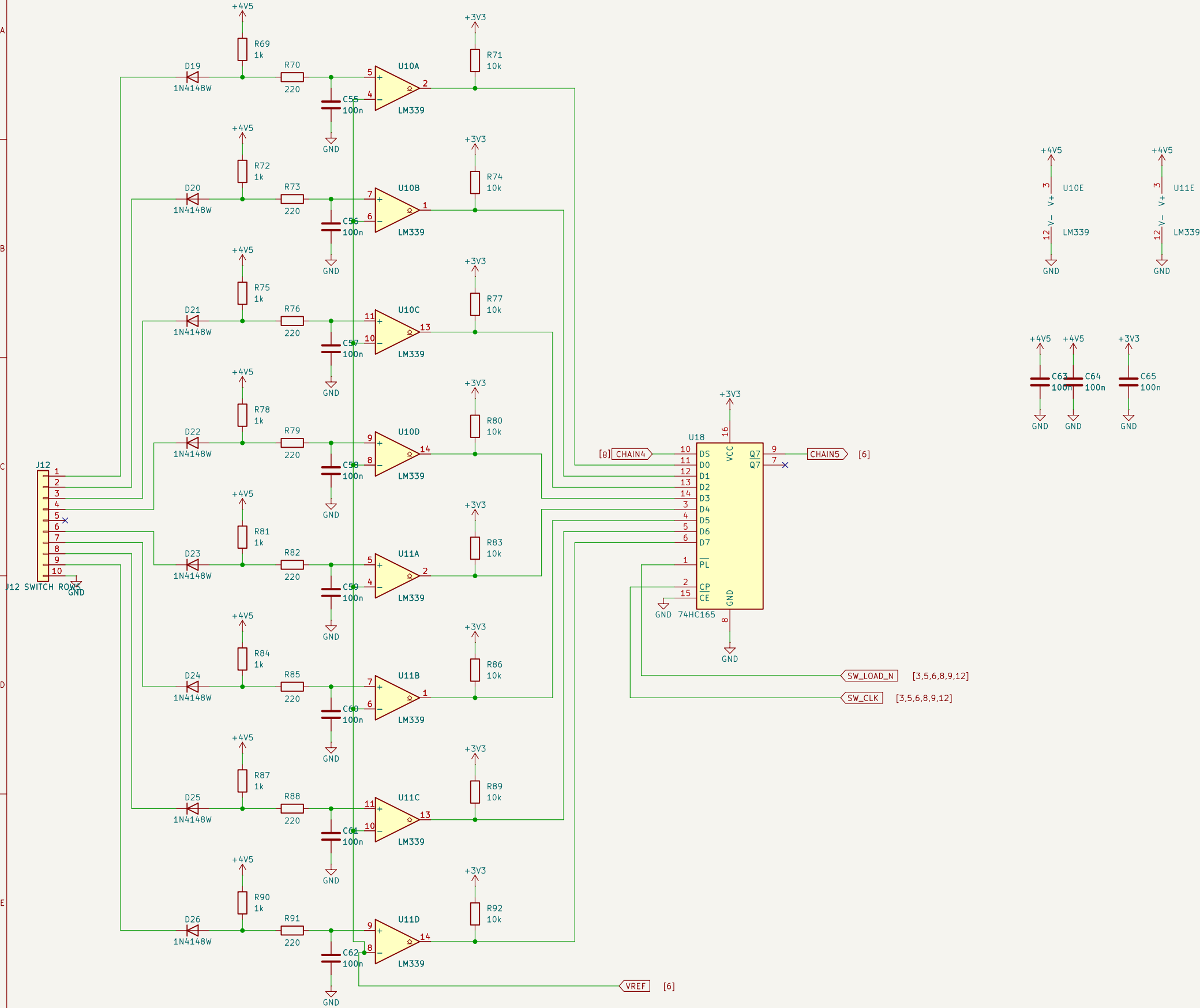
U20 latch shares SW_LOAD_N with the 74HC165 load (one pulse: 165 loads while low, 595 latches on the rising edge), U20 now follows U31 (DMD panels sheet), which carries the status LED and GI_SPARE.
One shift clock for both registers: PIO1 shifts the next strobe pattern into the 595 while the 74HC165 chain shifts in (ARCHITECTURE.md 5.3–5.4).
Strobes: 595 output high -> MMBT3904 on -> strobe pulled low (active), with the original 1 k pull-up to +4.5 V and 100 nF.
J1 keeps the original pinout (key on pin 2). Strobes 5–8 are driven even where a game does not use them (Q15).



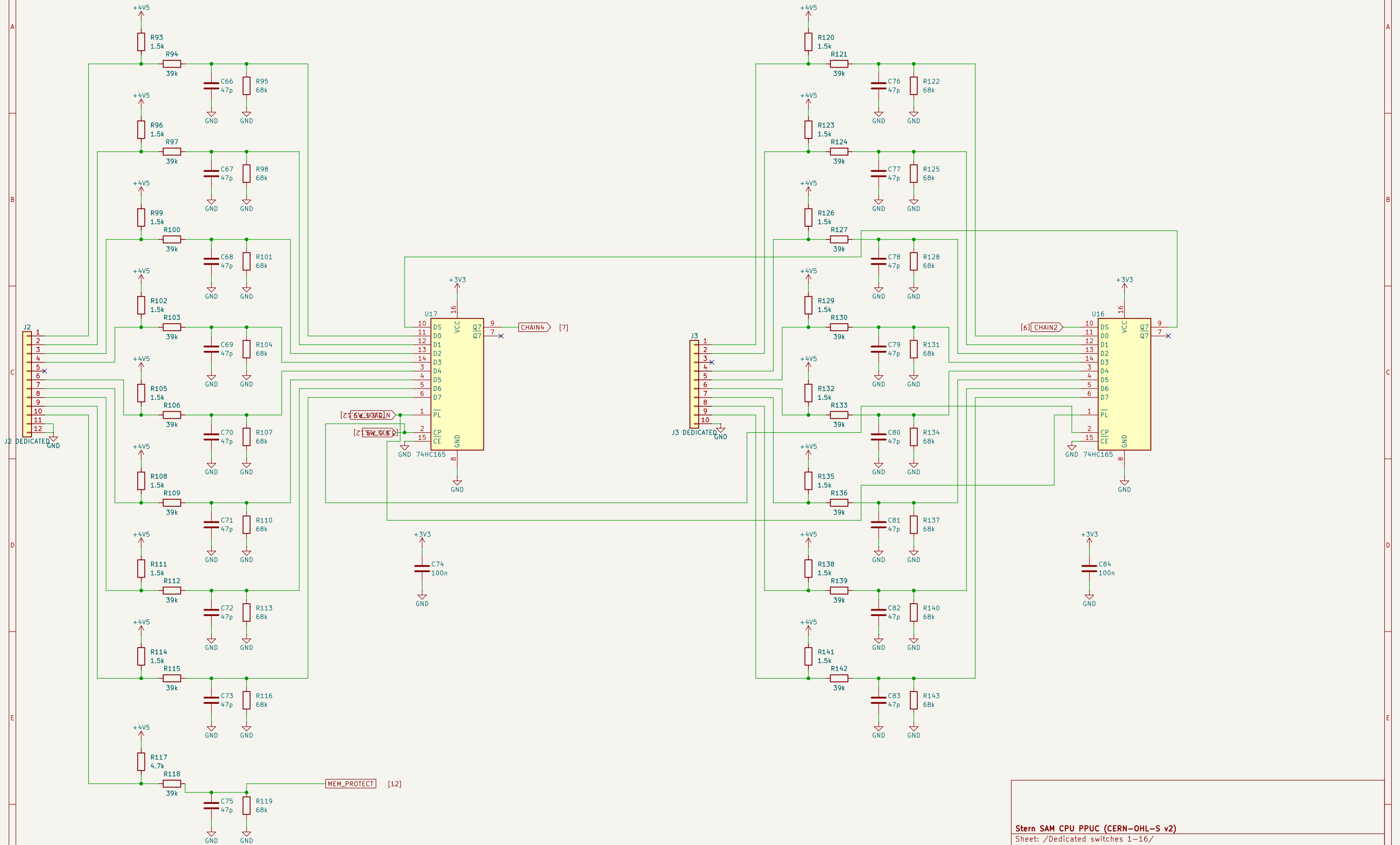
Per return (ARCHITECTURE.md 5.2–5.3): series diode from the connector, 1 k pull-up to +4.5 V, 220 R + 100 nF, LM339 against VREF (2.25 V). Output 1 = open, 0 = closed, as the ROM expects. Connector pin order follows the original drawing: check it with a meter.

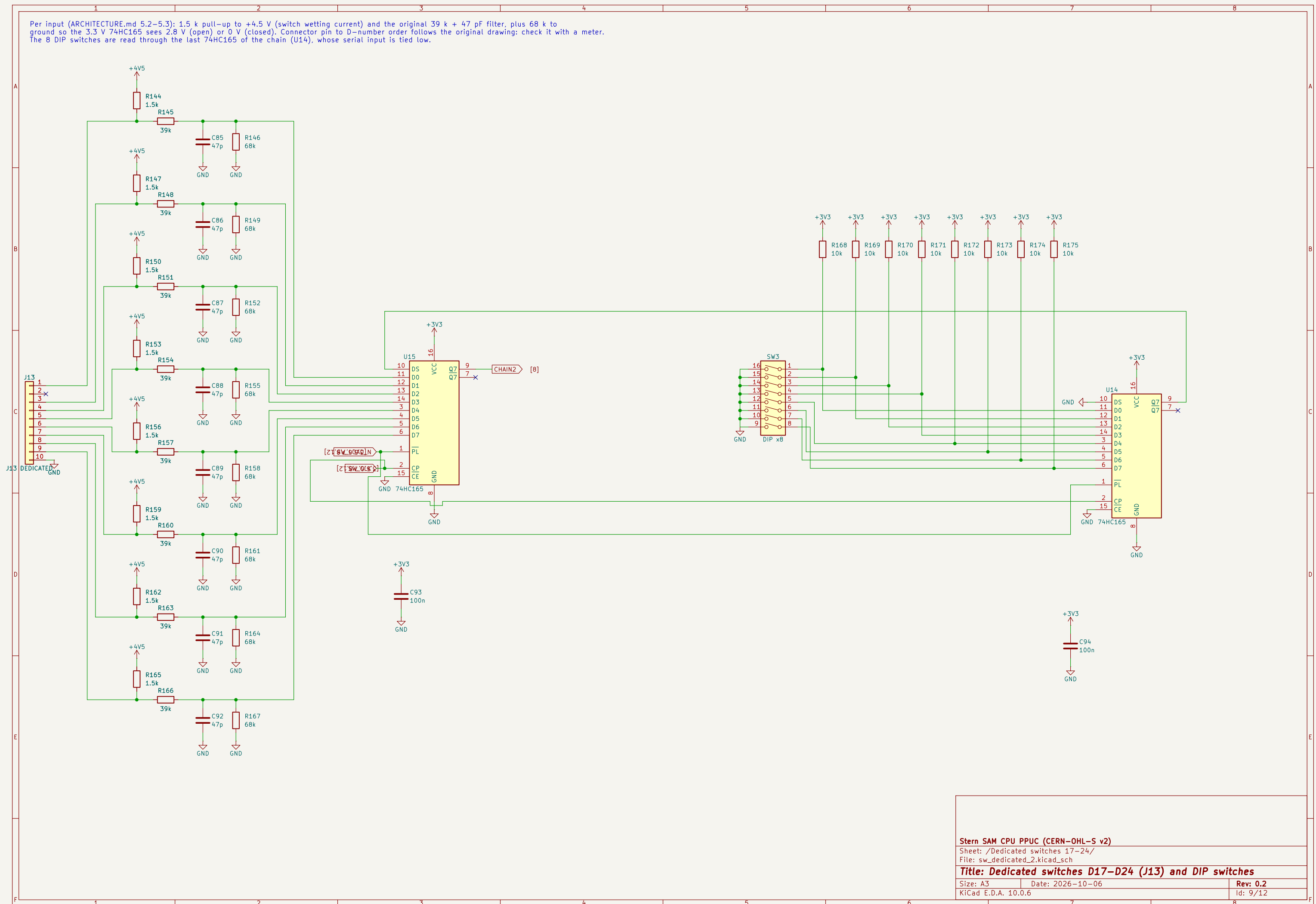


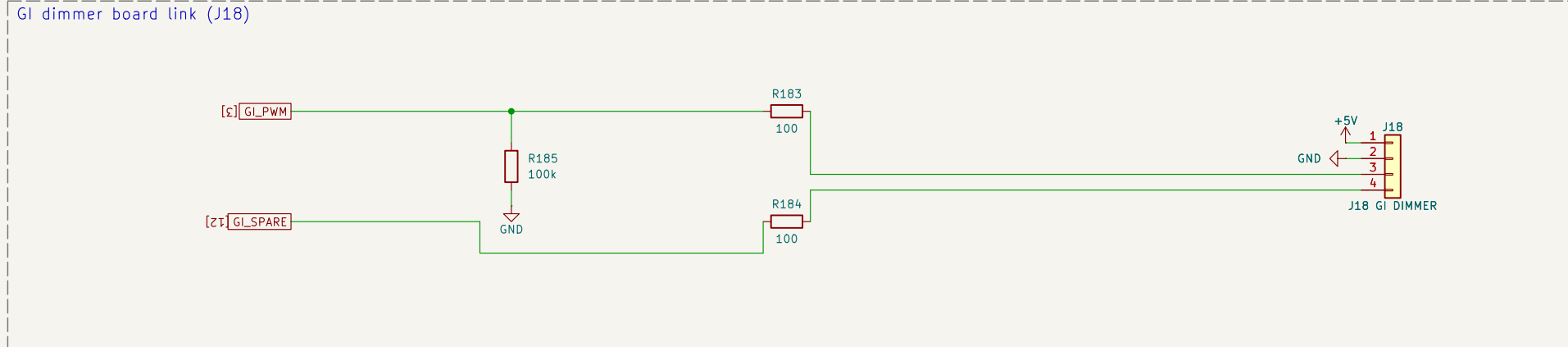
Per return (ARCHITECTURE.md 5.2–5.3): series diode from the connector, 1 k pull-up to +4.5 V, 220 R + 100 nF, LM339 against VREF (2.25 V). Output 1 = open, 0 = closed, as the ROM expects. Connector pin order follows the original drawing: check it with a meter.



Per input (ARCHITECTURE.md 5.2–5.3): 1.5 k pull-up to +4.5 V (switch wetting current) and the original 39 k + 47 pF filter, plus 68 k to ground so the 3.3 V 74HC165 sees 2.8 V (open) or 0 V (closed). Connector pin to D-number order follows the original drawing: check it with a meter. Memory protect (J2 pin 10) uses the same filter with the original 4.7 k pull-up and goes to input D0 of the 7th 74HC165 (U30, DMD panels sheet).

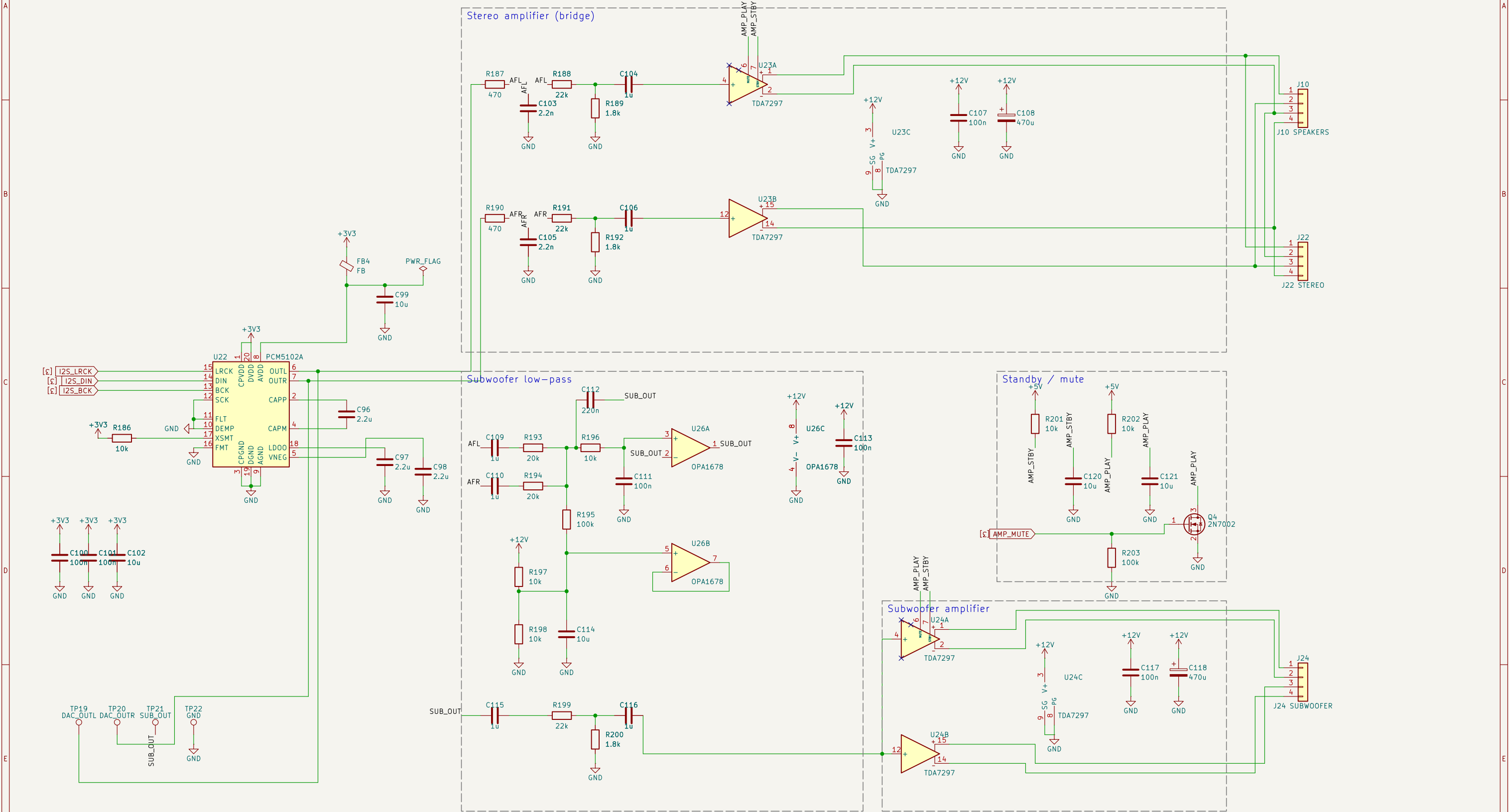






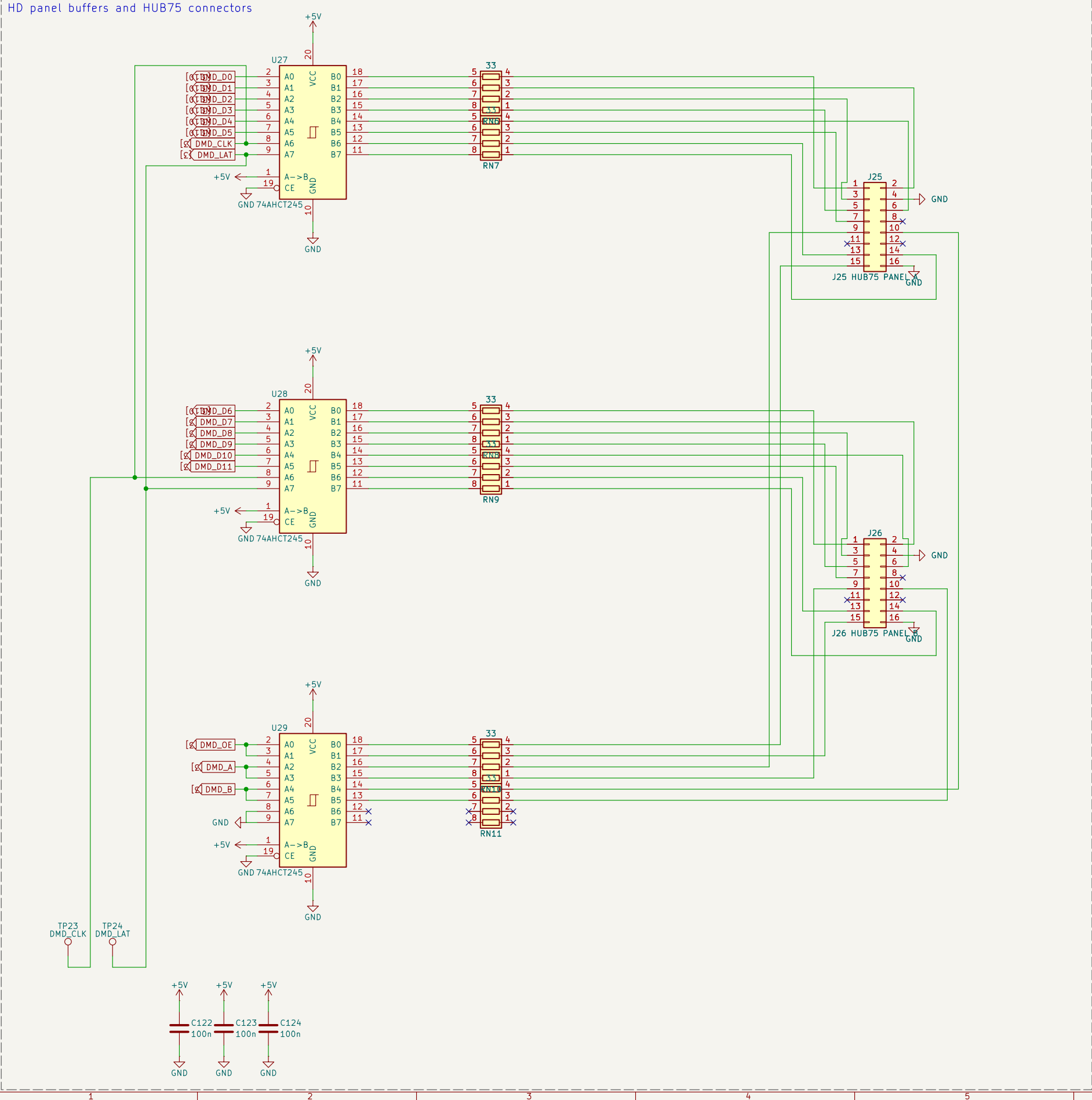
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Single +12 V supply (J23). U23 drives left / right in bridge (BTL): no output capacitor, but no speaker wire may go to ground.
J10 keeps the original connector: pins 1 / 2 = left + / right +, pins 3 / 4 (ground on the original) = left - / right -. Check the harness keeps the two speaker returns separate.
J22 = the same two channels as pairs (L+ L- R+ R-). Subwoofer: (L + R) / 2 through a 110 Hz 2nd-order low-pass (U26). U24 drives two bridged outputs on J24 (dual voice coil or two subwoofers).
TDA7297 gain is fixed at 32 dB: 22k / 1.8k input dividers give full output at DAC full scale. AMP_MUTE (RP2354B GPIO37) high mutes all outputs. U23 / U24 need a heatsink.



Two HUB75 panels (J25 = panel A, J26 = panel B) on 17 RP2354B lines (PIO2): 6 colour bits per panel (R1 G1 B1 R2 G2 B2), shared CLK, LAT, OE and 2 row-address lines (A, B). 74AHCT245 buffers on +5 V (HUB75 is 5 V logic), 33 R series resistors. Panel A data = DMD_D0–D5 (GPIO20–25), panel B = DMD_D6–D11 (GPIO26–31). DMD_D0–D6 also drive the original DMD buffer (J5): only one display path runs at a time. HUB75 C, D, E (pins 11, 12, 8) are not connected: panels with 4–5 row lines need 3 more GPIOs. U31: first 74HC595 of the strobe chain (status LED, GL_SPARE, 6 spare outputs). U30: last 74HC165 of the switch chain (coin door memory protect + 7 spare inputs). Both share SW_CLK and SW_LOAD_N (165 load / 595 latch on one pulse).

HD panel buffers and HUB75 connectors



Switch chain extensions

